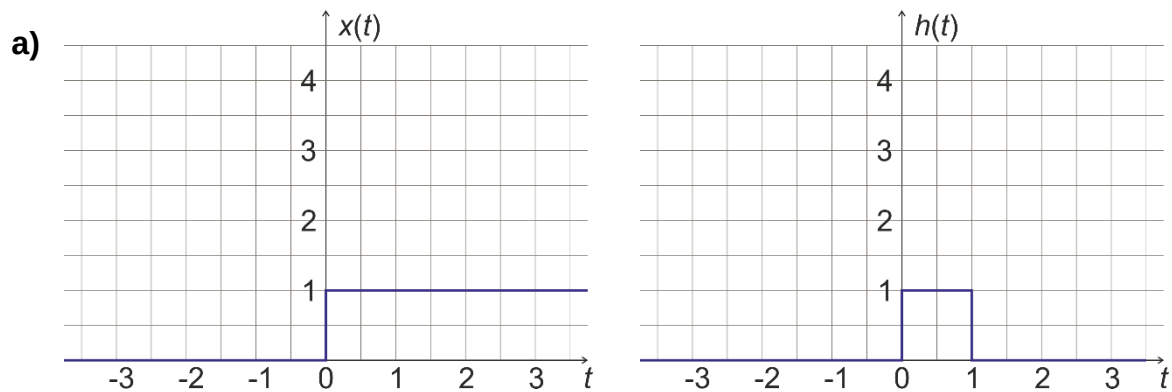


# Signal Processing

## Exercise Sheet 4 – Convolution - Solution

Summer Term 2026

### Question 1



b)

Mirror  $x(t)$

- Case 1:  $t \leq 0$   
 $y_1(t) = 0$ , as there exists no  $t \leq 0$  for which both functions are not equal to 0.
- Case 2:  $0 < t \leq 1$

$$\begin{aligned} y_2(t) &= \int_0^t x(t-\tau) \cdot h(\tau) d\tau = \int_0^t \sigma(t-\tau) \cdot \text{rect}(\tau - 0.5) d\tau \\ &= \int_0^t 1 \cdot 1 d\tau = [\tau]_0^t = t \end{aligned}$$

- Case 3:  $t > 1$

$$\begin{aligned} y_3(t) &= \int_0^1 x(t-\tau) \cdot h(\tau) d\tau = \int_0^1 \sigma(t-\tau) \cdot \text{rect}(\tau - 0.5) d\tau \\ &= \int_0^1 1 \cdot 1 d\tau = [\tau]_0^1 = 1 \end{aligned}$$

Mirror  $h(t)$

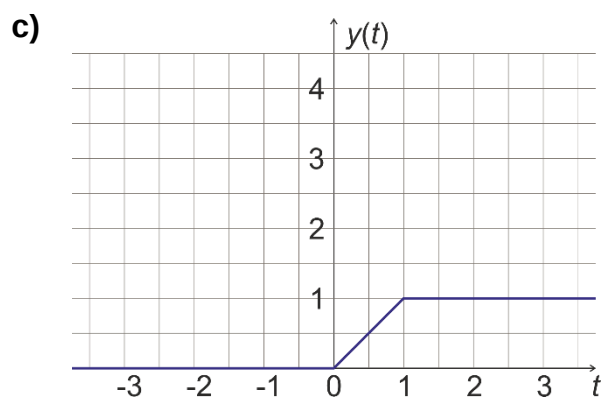
- Case 1:  $t \leq 0$   
 $y_1(t) = 0$ , as there exists no  $t \leq 0$  for which both functions are not equal to 0.
- Case 2:  $0 < t \leq 1$

$$\begin{aligned} y_2(t) &= \int_0^t x(\tau) \cdot h(t-\tau) d\tau = \int_0^t \sigma(\tau) \cdot \text{rect}((t-\tau) - 0.5) d\tau \\ &= \int_0^t 1 \cdot 1 d\tau = [\tau]_0^t = t \end{aligned}$$

- Case 3:  $t > 1$

$$\begin{aligned} y_3(t) &= \int_{t-1}^t x(\tau) \cdot h(t-\tau) d\tau = \int_{t-1}^t \sigma(\tau) \cdot \text{rect}((t-\tau) - 0,5) d\tau \\ &= \int_{t-1}^t 1 \cdot 1 d\tau = [\tau]_{t-1}^t = t - (t-1) = 1 \end{aligned}$$

$$y(t) = \begin{cases} 0 & t \leq 0 \\ t & 0 < t \leq 1 \\ 1 & t > 1 \end{cases}$$



## Question 2

### Mirror $x(t)$

- Case 1:  $t \leq 0$   
 $y_1(t) = 0$ , as there exists no  $t \leq 0$  for which both functions are not equal to 0.
- Case 2:  $0 < t \leq 2$   

$$y_2(t) = \int_0^t x(t-\tau) \cdot h(\tau) d\tau = \int_0^t (t-\tau) \cdot \text{rect}\left(\frac{t-\tau-1}{2}\right) \cdot \alpha \cdot \text{rect}\left(\frac{\tau-1.5}{3}\right) d\tau$$

$$= \int_0^t (t-\tau) \cdot \alpha d\tau = \alpha \cdot \left[t\tau - \frac{1}{2}\tau^2\right]_0^t = \frac{\alpha}{2}t^2$$
- Case 3:  $2 < t \leq 3$   

$$y_3(t) = \int_{t-2}^t (t-\tau) \cdot \alpha d\tau = \alpha \cdot \left[t\tau - \frac{1}{2}\tau^2\right]_{t-2}^t = 2\alpha$$
- Case 4:  $3 < t \leq 5$   

$$y_4(t) = \int_{t-2}^3 (t-\tau) \cdot \alpha d\tau = \alpha \cdot \left[t\tau - \frac{1}{2}\tau^2\right]_{t-2}^3 = \alpha \left(-\frac{1}{2}t^2 + 3t - 2.5\right)$$
- Case 5:  $t > 5$   
 $y_5(t) = 0$ , as there exists no  $t > 5$  for which both functions are not equal to 0.

### Mirror $h(t)$

- Case 1:  $t \leq 0$   
 $y_1(t) = 0$ , as there exists no  $t \leq 0$  for which both functions are not equal to 0.
- Case 2:  $0 < t \leq 2$   

$$y_2(t) = \int_0^t x(\tau) \cdot h(t-\tau) d\tau = \int_0^t \tau \cdot \text{rect}\left(\frac{\tau-1}{2}\right) \cdot \alpha \cdot \text{rect}\left(\frac{t-\tau-1.5}{3}\right) d\tau$$

$$= \int_0^t \tau \cdot \alpha d\tau = \alpha \cdot \left[\frac{1}{2}\tau^2\right]_0^t = \frac{\alpha}{2}t^2$$
- Case 3:  $2 < t \leq 3$   

$$y_3(t) = \int_0^2 \tau \cdot \alpha d\tau = \alpha \cdot \left[\frac{1}{2}\tau^2\right]_0^2 = 2\alpha$$
- Case 4:  $3 < t \leq 5$   

$$y_4(t) = \int_{t-3}^2 \tau \cdot \alpha d\tau = \alpha \cdot \left[\frac{1}{2}\tau^2\right]_{t-3}^2 = \alpha \left(-\frac{1}{2}t^2 + 3t - 2.5\right)$$
- Case 5:  $t > 5$   
 $y_5(t) = 0$ , as there exists no  $t > 5$  for which both functions are not equal to 0.

### Question 3

- 1 – D
- 2 – A
- 3 – F
- 4 – B
- 5 – C
- 6 – E

### Question 4

